

BILKENT UNIVERSITY
ELECTRICAL AND ELECTRONICS ENGINEERING

MATH 241
ENGINEERING MATHEMATICS I

MIDTERM EXAM II

8-12-2012

Instructions:

SOLUTION

- There are six questions in the exam. They have different weights. Solve all.
- Duration of the exam is 150 minutes.
- The exam is closed book, closed notes.
- No calculators are allowed.
- Please read each question carefully, explain your answers clearly, and show all your work to receive full credit.
- Boldface capital letters are matrices and boldface small letters are vectors.
- Do not detach pages.

Surname:.....

Name:.....

ID:.....

Section:.....

Signature:.....

Question	Grade
1	
2	
3	
4	
5	
6	
Total:	

Q1. (15 points) Let $A = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$

which has the reduced row echelon form given by $E = \begin{bmatrix} \textcircled{1} & 1 & 0 & 0 & 1 \\ 0 & 0 & \textcircled{1} & 0 & 1 \\ 0 & 0 & 0 & \textcircled{1} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.

- Find a basis for the row space of A .
- Find a basis for the column space of A .
- Find a basis for the null space of A .

(a) A basis for the row space of $A = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$

(It is also possible to select the corresponding rows from A)
 $\left\{ \begin{bmatrix} 2 \\ 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 2 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} \right\}$ is also a basis for $\text{Row}(A)$
 This result must be justified however, because rows may have been interchanged during E.R.O.s.

(b) A basis for the column space of $A = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

(c) $Ax = 0 \xrightarrow{\text{EROs}} Ex = 0 \Rightarrow x_4 = 0 \text{ let } x_5 = \alpha, x_2 = \beta$
 $\Rightarrow x_3 = -\alpha \quad x_1 = -\alpha - \beta$

A basis for the null space of $A = \left\{ \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}$

Q2. (10 points) Let $A_{m \times n}$ and $B_{n \times m}$ be such that both $I_m + AB$ and $I_n + BA$ are invertible.

Show that $(I_m + AB)^{-1} A = A(I_n + BA)^{-1}$ where I_m is the $m \times m$ identity matrix and I_n is the $n \times n$ identity matrix.

$$\left(\underline{I}_m + \underline{A} \underline{B} \right)^{-1} \underline{A} = \underline{A} \left(\underline{I}_n + \underline{B} \underline{A} \right)^{-1}$$

$$\Rightarrow \underline{A} = \left(\underline{I}_m + \underline{A} \underline{B} \right) \underline{A} \left(\underline{I}_n + \underline{B} \underline{A} \right)^{-1}$$

$$\Rightarrow \underline{A} = \underline{A} + \underline{A} \underline{B} \underline{A} \left(\underline{I}_n + \underline{B} \underline{A} \right)^{-1}$$

$$\Rightarrow \underline{A} = \underline{A} \underbrace{\left(\underline{I}_n + \underline{B} \underline{A} \right) \left(\underline{I}_n + \underline{B} \underline{A} \right)^{-1}}_{\underline{I}}$$

$$\Rightarrow \underline{A} = \underline{A} \quad \checkmark$$

OR

$$\left(\underline{I}_m + \underline{A} \underline{B} \right)^{-1} \underline{A} = \underline{A} \left(\underline{I}_n + \underline{B} \underline{A} \right)^{-1}$$

$$\Rightarrow \left(\underline{I}_m + \underline{A} \underline{B} \right)^{-1} \underline{A} \left(\underline{I}_n + \underline{B} \underline{A} \right) = \underline{A}$$

$$\Rightarrow \left(\underline{I}_m + \underline{A} \underline{B} \right)^{-1} \left(\underline{A} + \underline{A} \underline{B} \underline{A} \right) = \underline{A}$$

$$\Rightarrow \underbrace{\left(\underline{I}_m + \underline{A} \underline{B} \right)^{-1} \left(\underline{I}_m + \underline{A} \underline{B} \right)}_{\underline{I}} \underline{A} = \underline{A}$$

$$\Rightarrow \underline{A} = \underline{A}$$

Q3. (15 points) The 3×3 matrix A reduces to the identity matrix I by the following three elementary row operations in order:

(1) $-4\text{Row}_1 + \text{Row}_2 \rightarrow \text{Row}_2$

(2) $-3\text{Row}_1 + \text{Row}_3 \rightarrow \text{Row}_3$

(3) $-\text{Row}_3 + \text{Row}_2 \rightarrow \text{Row}_2$

a) Compute A^{-1} ,

b) Find A . (Note that check your result via $A^{-1}A = AA^{-1} = I$. Otherwise no partial credit will be given in case of wrong answer)

$$\left[\begin{array}{c|c} A & I \end{array} \right] \xrightarrow{-4R_1 + R_2 \rightarrow R_2} \left[\begin{array}{c|c} & \end{array} \right] \xrightarrow{-3R_1 + R_3 \rightarrow R_3} \left[\begin{array}{c|c} & \end{array} \right] \xrightarrow{-R_3 + R_2 \rightarrow R_2} \left[\begin{array}{c|c} I & A^{-1} \end{array} \right]$$

(a)
$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -4 & 1 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right] \xrightarrow{-4R_1 + R_2 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ -4 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -3 & 0 & 1 \end{array} \right] \xrightarrow{-3R_1 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ -4 & 1 & 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 3 & 0 & 1 \end{array} \right] \xrightarrow{-R_3 + R_2 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & -1 & 4 & 1 & -1 \\ -3 & 0 & 1 & 3 & 0 & 1 \end{array} \right] = A^{-1}$$

(b) You can apply EROs in the reverse order and obtain A from I

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3 + R_2 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{3R_1 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 3 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{4R_1 + R_2 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 4 & 1 & 1 & 0 & 0 & 0 \\ 3 & 0 & 1 & 0 & 0 & 1 \end{array} \right] = A$$

OR (alternative way)

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \xrightarrow[-3R_1 + R_3 \rightarrow R_3]{-4R_1 + R_2 \rightarrow R_2} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} - 4a_{11} & a_{22} - 4a_{12} & a_{23} - 4a_{13} \\ a_{31} - 3a_{11} & a_{32} - 3a_{12} & a_{33} - 3a_{13} \end{bmatrix}$$

$$\xrightarrow{-R_3 + R_2 \rightarrow R_2} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} - a_{11} - a_{31} & a_{22} - a_{12} - a_{32} & a_{23} - a_{13} - a_{33} \\ a_{31} - 3a_{11} & a_{32} - 3a_{12} & a_{33} - 3a_{13} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \boxed{a_{11} = 1}, \boxed{a_{12} = 0}, \boxed{a_{13} = 0}, \quad a_{31} - 3a_{11} = 0 \Rightarrow \boxed{a_{31} = 3}, \quad a_{32} - 3a_{12} = 0 \Rightarrow \boxed{a_{32} = 0},$$

$$a_{33} - 3a_{13} = 1 \Rightarrow \boxed{a_{33} = 1}, \quad a_{21} - a_{11} - a_{31} = 0 \Rightarrow \boxed{a_{21} = 4}, \quad a_{22} - a_{12} - a_{32} = 1 \Rightarrow \boxed{a_{22} = 1},$$

$$a_{23} - a_{13} - a_{33} = 0 \Rightarrow \boxed{a_{23} = 1}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 1 \\ 3 & 0 & 1 \end{bmatrix} //$$

OR take the inverse of A^{-1}

Q4. Let $A_{3 \times 3}$ be a real and symmetric matrix, and $(0, v_1)$, $(-1, v_2)$, and $(1, v_3)$ be its eigenpairs.

(a) (5 points) Determine a basis for the null space of A in terms of v_1 , v_2 , and v_3 .

(b) (5 points) Determine a basis for the row space of A in terms of v_1 , v_2 , and v_3 .

(c) (5 points) Determine a basis for the column space of A in terms of v_1 , v_2 , and v_3 .

In each part you have to justify your answer to receive full credit.

Because $\underline{A}$ is real and symmetric, and all eigenvalues are distinct, $\{v_1, v_2, v_3\}$ forms an orthogonal set.

$$(a) \quad \underline{A} v_1 = 0 v_1 = \underline{0} \Rightarrow v_1 \in \text{Null}(\underline{A})$$

We must check if another LI vector (say u) can be also in the null space of $\underline{A}$.

However, let's first do part (b)

$$(b) \quad \text{Row}(\underline{A}) \perp \text{Null}(\underline{A})$$

$$\left(\text{Proof: } \underline{A} = \begin{bmatrix} \underline{r}_1^T \\ \vdots \\ \underline{r}_n^T \end{bmatrix} \underline{x} = \underline{0} \Rightarrow \underline{x} \in \text{Null}(\underline{A}) \Rightarrow \underline{r}_i^T \underline{x} = 0 \Rightarrow \text{span}\{\underline{r}_1, \underline{r}_2, \dots, \underline{r}_n\} \text{ is orthogonal to } \underline{x} \right)$$

$$\text{If } v_1 \in \text{Null}(\underline{A}) \Rightarrow \{v_2, v_3\} \in \text{Row}(\underline{A}) \text{ and } \{v_2, v_3\} \text{ is a LI set } (v_2 \text{ and } v_3 \text{ are orthogonal})$$

$$\text{Because } \underbrace{\dim(\text{Row}(\underline{A}))}_2 + \underbrace{\dim(\text{Null}(\underline{A}))}_1 = 3$$

$$\therefore \text{A basis for } \text{Row}(\underline{A}) = \{v_2, v_3\}$$

$$\text{A basis for } \text{Null}(\underline{A}) = \{v_1\}$$

$$(c) \quad \text{From } \underline{A} \underline{x} = \lambda \underline{x} \quad (-1, v_2) \text{ and } (1, v_3) \text{ are eigenpairs}$$

$$\Rightarrow \underline{A} v_2 = -v_2 \Rightarrow v_2 \in \text{Col}(\underline{A})$$

$$\underline{A} v_3 = v_3 \Rightarrow v_3 \in \text{Col}(\underline{A})$$

$$\dim(\text{Row}(\underline{A})) = \dim(\text{Col}(\underline{A})) = 2$$

$$\text{because } \{v_2, v_3\} \text{ is a LI set } (v_2 \text{ and } v_3 \text{ are orthogonal})$$

$$\Rightarrow \text{A basis for } \text{Col}(\underline{A}) = \{v_2, v_3\}$$

$$\begin{aligned} \text{OR} \\ \underline{A} &= \underline{A}^T \\ \Rightarrow \text{Row}(\underline{A}) &= \text{Col}(\underline{A}) \end{aligned}$$

Q5. Let $A_{3 \times 3} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$

(a) (15 points) Determine the eigenvalues and a basis for the associated eigenspaces of A .

(b) (15 points) Calculate $A^{101} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$.

$$(a) \det(A - \lambda I) = \begin{vmatrix} 2-\lambda & -1 & -1 \\ -1 & 2-\lambda & -1 \\ -1 & -1 & 2-\lambda \end{vmatrix} = (2-\lambda) \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} + 1 \begin{vmatrix} -1 & -1 \\ -1 & 2-\lambda \end{vmatrix} - 1 \begin{vmatrix} -1 & 2-\lambda \\ -1 & -1 \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda) [(2-\lambda)^2 - 1] + (\lambda - 2 - 1) - (1 + 2 - \lambda) = (2-\lambda) [\lambda^2 - 4\lambda + 3] + 2(\lambda - 3) = 0$$

$$\Rightarrow 2\lambda^2 - 8\lambda + 6 - \lambda^3 + 4\lambda^2 - 3\lambda + 2\lambda - 6 = -\lambda^3 + 6\lambda^2 - 9\lambda = \lambda(\lambda^2 - 6\lambda + 9) = 0$$

$$\Rightarrow \lambda_1 = 0 \quad \text{and} \quad \lambda_2 = 3 \quad \text{with} \quad k=2 \quad (k: \text{multiplicity})$$

* For $\lambda_1 = 0$

$$(A - \lambda_1 I) \underline{v} = \underline{0} \Rightarrow \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{\substack{-R_1 + R_2 \rightarrow R_2 \\ 2R_1 + R_3 \rightarrow R_3}} \begin{bmatrix} -1 & -1 & 2 \\ 0 & 3 & -3 \\ 0 & -3 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\xrightarrow{R_2 + R_3 \rightarrow R_3} \begin{bmatrix} -1 & -1 & 2 \\ 0 & 3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{let } v_3 = \alpha \Rightarrow v_2 = \alpha \Rightarrow -v_1 = v_2 - 2v_3 \Rightarrow v_1 = \alpha$$

$$\Rightarrow \underline{v} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}. \quad \text{Thus, a basis for the associated eigenspace is } \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

* for $\lambda_2 = 3$

$$(A - \lambda_2 I) \underline{x} = \underline{0} \Rightarrow \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \xrightarrow{\substack{-R_1 + R_2 \rightarrow R_2 \\ -R_1 + R_3 \rightarrow R_3}} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \text{let } x_3 = \alpha, x_2 = \beta$$

$$\Rightarrow \underline{x} = \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + \beta \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}. \quad \text{Thus, a basis for the associated eigenspace is } \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

(b) This part can be done in many ways. Some are short, some are longer!

(i) You may use diagonalization. (LONG WAY)

$\underline{A}$ can be diagonalized since it has full set of eigenvectors.

Let $\underline{Q} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ and $\underline{\Lambda} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, find $\underline{Q}^{-1}$

$$[\underline{Q} | \underline{I}] \xrightarrow{\text{ERO}} [\underline{I} | \underline{Q}^{-1}] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{-R_1+R_2 \rightarrow R_2 \\ -R_1+R_3 \rightarrow R_3}} \left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{\substack{-R_2/2 \rightarrow R_2 \\ \rightarrow R_3}} \left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 3/2 & -1/2 & -1/2 & 1 \end{array} \right] \xrightarrow{+2/3 R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1/3 & -1/3 & 2/3 \end{array} \right] \xrightarrow{-R_3+R_2 \rightarrow R_2}$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 2 & 0 & -4/3 & 4/3 & -2/3 \\ 0 & 0 & 1 & -1/3 & -1/3 & 2/3 \end{array} \right] \xrightarrow{R_2/2 \rightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & -1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2/3 & 2/3 & -1/3 \\ 0 & 0 & 1 & -1/3 & -1/3 & 2/3 \end{array} \right] \xrightarrow{\substack{R_3+R_1 \rightarrow R_1 \\ R_2+R_1 \rightarrow R_1}}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 1 & 0 & -1/3 & 2/3 & -1/3 \\ 0 & 0 & 1 & -1/3 & -1/3 & 2/3 \end{array} \right] \Rightarrow \underline{Q}^{-1} = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix}$$

It is also possible to orthonormalize & so that $\underline{Q}^T = \underline{Q}^{-1}$. Gram-Schmidt may be used. See *

$$\Rightarrow \underline{Q}^{-1} \underline{A} \underline{Q} = \underline{\Lambda} \Rightarrow \underline{Q}^{-1} \underline{A}^{101} \underline{Q} = \underline{\Lambda}^{101} \Rightarrow \underline{A}^{101} = \underline{Q} \underline{\Lambda}^{101} \underline{Q}^{-1}$$

$$\Rightarrow \underline{A}^{101} = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3^{101} & 0 \\ 0 & 0 & 3^{101} \end{bmatrix} \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} = 3^{101} \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} //$$

$$\Rightarrow \underline{A}^{101} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 3^{101} \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} //$$

(ii) Use the identity $\underline{A}^k \underline{x} = \lambda^k \underline{x}$ provided that $(\lambda, \underline{x})$ is an eigenpair of $\underline{A}$

$$\begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \alpha_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \alpha_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + \alpha_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \left[\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 1 & 1 & 0 & 3 \\ 1 & 0 & 1 & 1 \end{array} \right] \xrightarrow{\substack{-R_1+R_2 \rightarrow R_2 \\ -R_1+R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & -1 \end{array} \right] \xleftrightarrow{R_2 \leftrightarrow R_3}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 2 & 1 & 1 \end{array} \right] \xrightarrow{-2R_2+R_3 \rightarrow R_3} \left[\begin{array}{ccc|c} 1 & -1 & -1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & -3 & 3 \end{array} \right] \Rightarrow \alpha_3 = -1, \alpha_2 = 1, \alpha_1 = 2$$

$$\Rightarrow \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \underline{A}^{101} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \underline{A}^{101} \left(2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right) = 2 \lambda_1^{101} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \lambda_2^{101} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - \lambda_3^{101} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow \underline{A}^{101} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 0 + 3^{101} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} - 3^{101} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = 3^{101} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} //$$

(iii) Use Cayley Hamilton Theorem to find $\underline{A}^{101}$

$$\underline{A}^{101} = \alpha_2 \underline{A}^2 + \alpha_1 \underline{A} + \alpha_0 \underline{I} \Rightarrow \lambda_1^{101} = \alpha_2 \lambda_1^2 + \alpha_1 \lambda_1 + \alpha_0 \quad (\text{for } \lambda_1 = 0)$$

$$\Rightarrow \lambda_2^{101} = \alpha_2 \lambda_2^2 + \alpha_1 \lambda_2 + \alpha_0 \quad (\text{for } \lambda_2 = 3)$$

$$\Rightarrow 101 \lambda_2^{100} = 2 \alpha_2 \lambda_2 + \alpha_1 \quad (\text{for } \lambda_2 = 3)$$

$$\boxed{\alpha_0 = 0}$$

$$\left. \begin{array}{l} 3^{101} = 9 \alpha_2 + 3 \alpha_1 \\ \alpha_1 3^{100} = 6 \alpha_2 + \alpha_1 \end{array} \right\} \Rightarrow \alpha_2 = \left(\frac{100}{9} \right) \times 3^{101}$$

$$4 \quad \alpha_1 = -33 \times 3^{101}$$

$$\underline{A}^2 = \underline{A} \underline{A} = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -3 & -3 \\ -3 & 6 & -3 \\ -3 & -3 & 6 \end{bmatrix}$$

As a matter of fact since $\underline{A}^2 = 3\underline{A}$, $\underline{A}^{101} = \alpha_1 \underline{A} + \alpha_0 \underline{I}$ can also be used. However this must be justified.

$$\Rightarrow \underline{A}^{101} = \left(\frac{100}{9} \right) \times 3^{101} \begin{bmatrix} 6 & -3 & -3 \\ -3 & 6 & -3 \\ -3 & -3 & 6 \end{bmatrix} - 3^{101} \times 33 \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} = 3^{101} \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} //$$

$$\Rightarrow \underline{A}^{101} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = 3^{101} \begin{bmatrix} 2/3 & -1/3 & -1/3 \\ -1/3 & 2/3 & -1/3 \\ -1/3 & -1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \times 3^{101} //$$

(iv) Finally just via multiplication (MAY NOT WORK always)

$$\underline{A}^{101} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \underline{A}^{100} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \underline{A}^{100} \begin{bmatrix} 0 \\ 3 \\ -3 \end{bmatrix} = \underline{A}^{99} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \times 3 = \underline{A}^{98} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} 3^2 //$$

$$= \underline{A}^{97} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} 3^3 = \dots = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \times 3^{100} = 3^{101} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} //$$

* Addendum to solution of Q5 a) i) using diagonalization
 The Q matrix given above $\left\{ Q = \begin{bmatrix} 1 & -1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \right\}$ is not orthonormal
 and therefore finding its inverse takes effort. Instead
 we may use an orthonormal Q .

The eigenspace basis for $\lambda=0$ is $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$. We can
 normalize this and take $\hat{e}_1 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$ $\uparrow$
 e_1

The eigenspace basis for $\lambda=3$ is $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$
 $\uparrow$ $\uparrow$
 e_2 e_3

Take $\hat{e}_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$.

$$\text{Then } \hat{e}_3 = \frac{e_3 - (e_3 \cdot \hat{e}_2) \hat{e}_2}{\|e_3 - (e_3 \cdot \hat{e}_2) \hat{e}_2\|} = \frac{\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}}{\text{denom}}$$

$$= \frac{\begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}}{\sqrt{\frac{1}{4} + \frac{1}{4} + 1}} = \sqrt{\frac{2}{3}} \begin{bmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$\text{Thus } Q = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix} \Rightarrow Q^{-1} = Q^T = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} \end{bmatrix}$$

$$A^{101} = Q \Lambda^{101} Q^T = Q \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3^{101} & 0 \\ 0 & 0 & 3^{10} \end{bmatrix} Q^T$$

$$A^{101} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3^{101} & 0 \\ 0 & 0 & 3^{10} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} 3^{101} \\ -\frac{1}{\sqrt{6}} 3^{101} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{2} 3^{101} + \frac{1}{\sqrt{6}} \frac{\sqrt{3}}{\sqrt{2}} 3^{101} \\ \frac{1}{2} 3^{101} + \frac{1}{\sqrt{6}} \frac{\sqrt{3}}{\sqrt{2}} 3^{101} \\ -\frac{2}{\sqrt{6}} \frac{\sqrt{3}}{\sqrt{2}} 3^{101} \end{bmatrix} = \begin{bmatrix} 0 \\ 3^{101} \\ -3^{101} \end{bmatrix}$$

Q6. For each of the following statements, provide a complete proof if the statement is true, otherwise provide a counter example.

- (a) (5 points) It is given that $Ax = b$ is consistent. If $A^T y = 0$ then y is always orthogonal to b .
- (b) (5 points) If a linear system of equations has no solution (i.e., $Ax = b$ is inconsistent), then the rows of its coefficient matrix are linearly dependent.
- (c) (5 points) All real skew symmetric matrices have $\lambda = 0$ as one of their eigenvalues.

(a) TRUE! $Ax = b$ consistent $\Rightarrow b \in \text{Col}(A)$

$$\text{Let } A = [a_1 \ a_2 \ \dots \ a_n] \Rightarrow A^T = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{bmatrix} \Rightarrow A^T y = 0 \Rightarrow a_i^T y = 0 \quad \forall \ 1 \leq i \leq n$$

$$\text{but } b \in \text{Col}(A) \Rightarrow b = \alpha_1 a_1 + \alpha_2 a_2 + \dots + \alpha_n a_n \\ \Rightarrow b^T y = \alpha_1 a_1^T y + \alpha_2 a_2^T y + \dots + \alpha_n a_n^T y = 0 \Rightarrow b \perp y$$

(b) TRUE! $Ax = b$ has no solution implies that Gaussian Elimination (i.e., a finite sequence of EROs) will end up with $E x = d$ where a full row of E is 0 and its corresponding row in d is not. Therefore, rows of E are linearly dependent because 0 can be obtained from the other rows. Because row space of A and E are the same, the same linear dependence is also true for the rows of A .

(c) FALSE! Counter example

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \text{ where } A^T = -A; \quad \det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ -1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 + 1 = 0 \Rightarrow \lambda = \pm j$$